

Hallucination and Constraint Satisfaction Report

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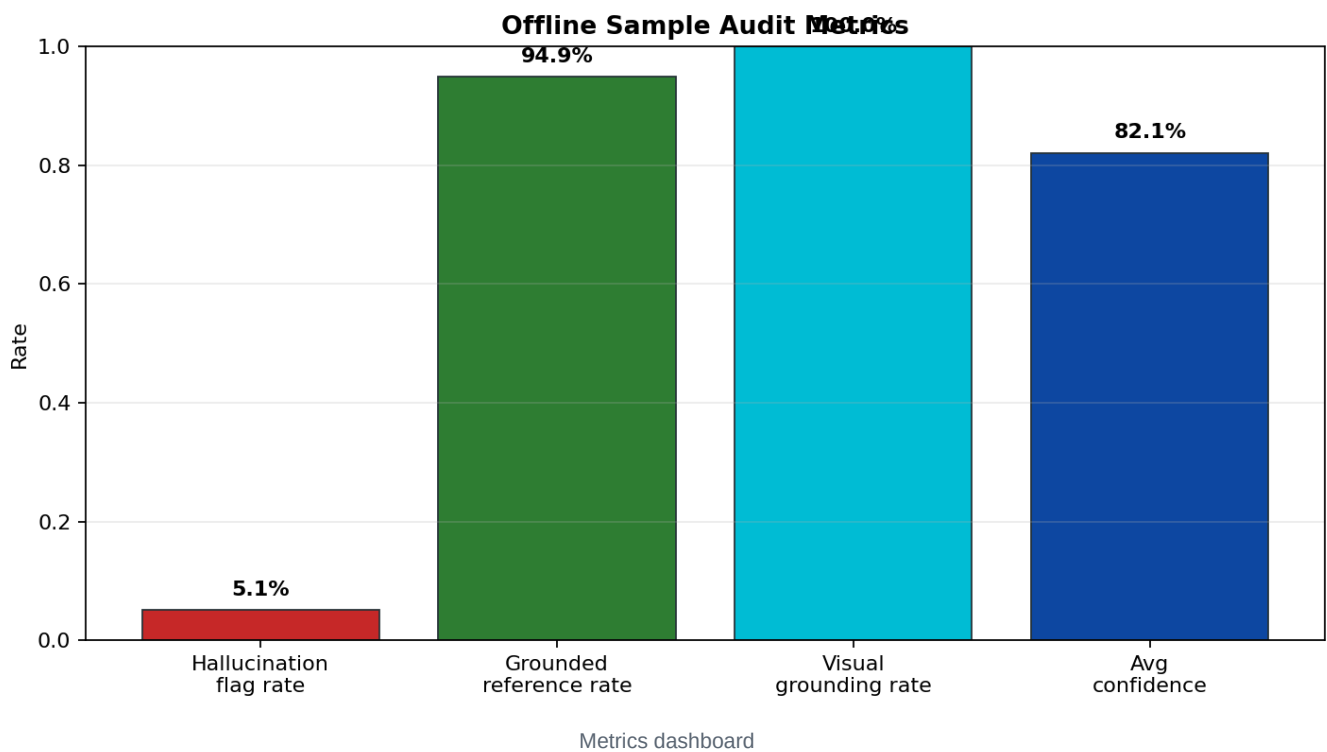
Scope: Offline sample audit for the ClinicGuard-ReportGen prototype.

Clinical status: Research and education only; not for diagnosis.

Summary

This report documents how ClinicGuard reduces unsupported claims and how the current repository reports hallucination-related metrics. The expanded evidence log contains 75 claim-level rows across 10 bundled sample cases.

Measure	Value
Generated claim rows	59
Protective refusals	16
Flagged hallucinated generated claims	3
Sample hallucination flag rate	5.1%
Grounded reference rate	94.9%
Visual grounding rate	100.0%
Average generated-claim confidence	82.1%



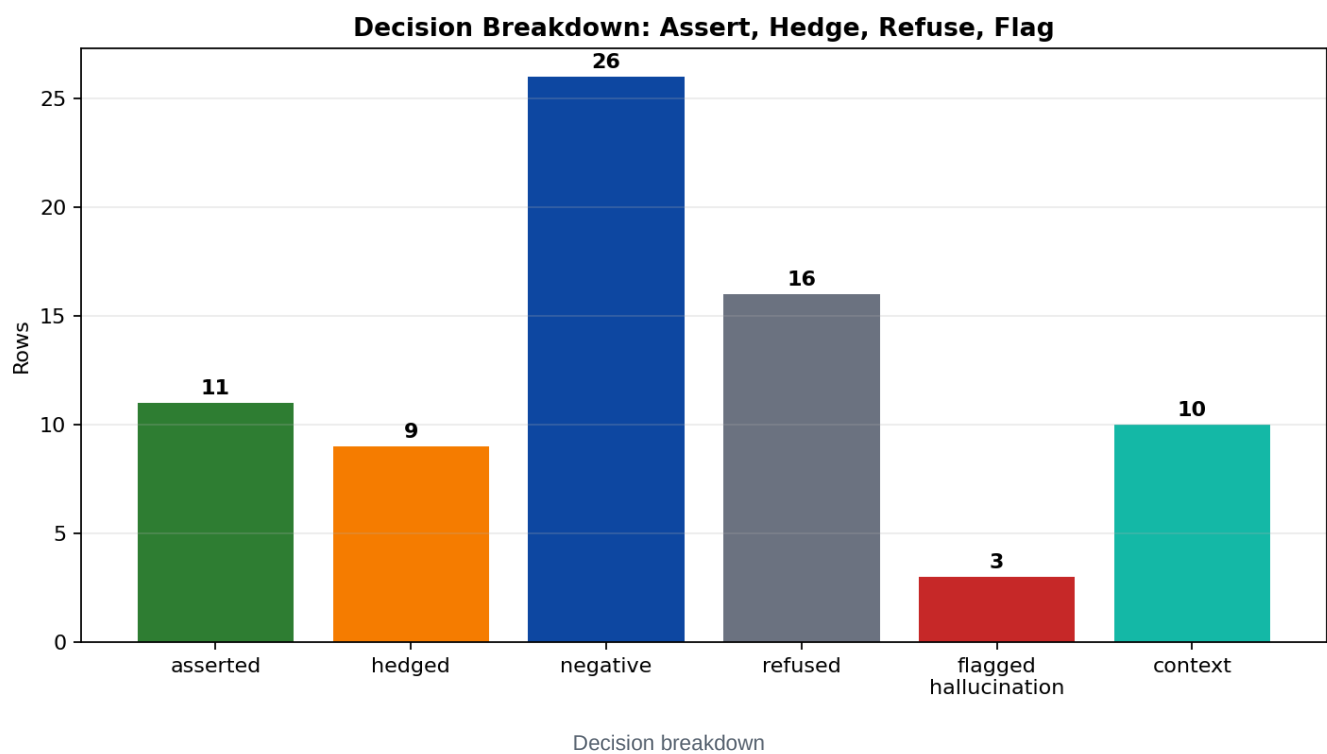
Definition

In this project, a hallucination is a generated clinical assertion that is not supported by visual evidence, patient history, or prior report text. A claim is flagged when it asserts a finding but the available sources are insufficient.

Constraint Satisfaction Strategy

ClinicGuard uses three safety controls:

- **Confidence-gated assertions:** Findings above 0.75 can be asserted.
- **Hedged uncertainty:** Findings between 0.50 and 0.75 are written with cautious language.
- **Protective refusal:** Findings below 0.50 are not asserted and are logged as refused/not generated.



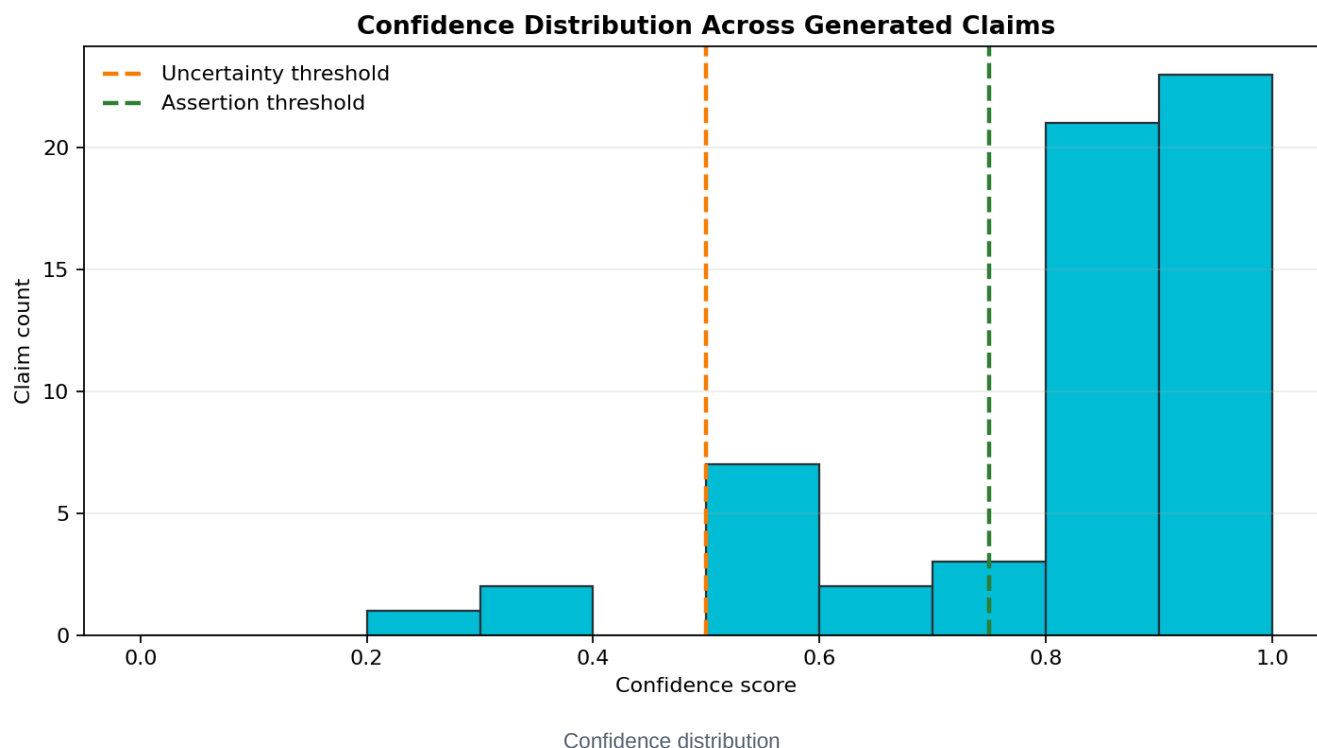
Evidence Log Schema

Column	Meaning
sample_id	Stable case identifier.
generated_claim	Claim text or refusal statement.
source_type	visual, history, prior, refusal_gate, or UNGROUNDED.
source_reference	Concrete source pointer such as bbox, global assessment, or input text reference.
confidence_score	Model or verifier confidence associated with the row.
hallucinated	Boolean flag for unsupported generated claims.
finding	Pathology or evidence category.
decision	asserted, hedged, negative, refused, context, or flagged_hallucination.
verification_note	Human-readable review note.

Flagged Claims

Sample	Flagged claim	Finding	Confidence	Review note
CASE-001	Large right upper lobe mass	Mass	0.21	Detector flags this as unsupported by visual, history, or prior evidence.
CASE-005	Definite pulmonary edema	Edema	0.37	Detector flags this as unsupported by visual, history, or prior evidence.
CASE-007	Moderate pleural effusion	Effusion	0.39	Detector flags this as unsupported by visual, history, or prior evidence.

Confidence and Refusal Behavior



Rows below the uncertainty threshold are not forced into the report. This is a meaningful safety behavior: refusing a low-evidence finding is better than producing a fluent but unsupported statement.

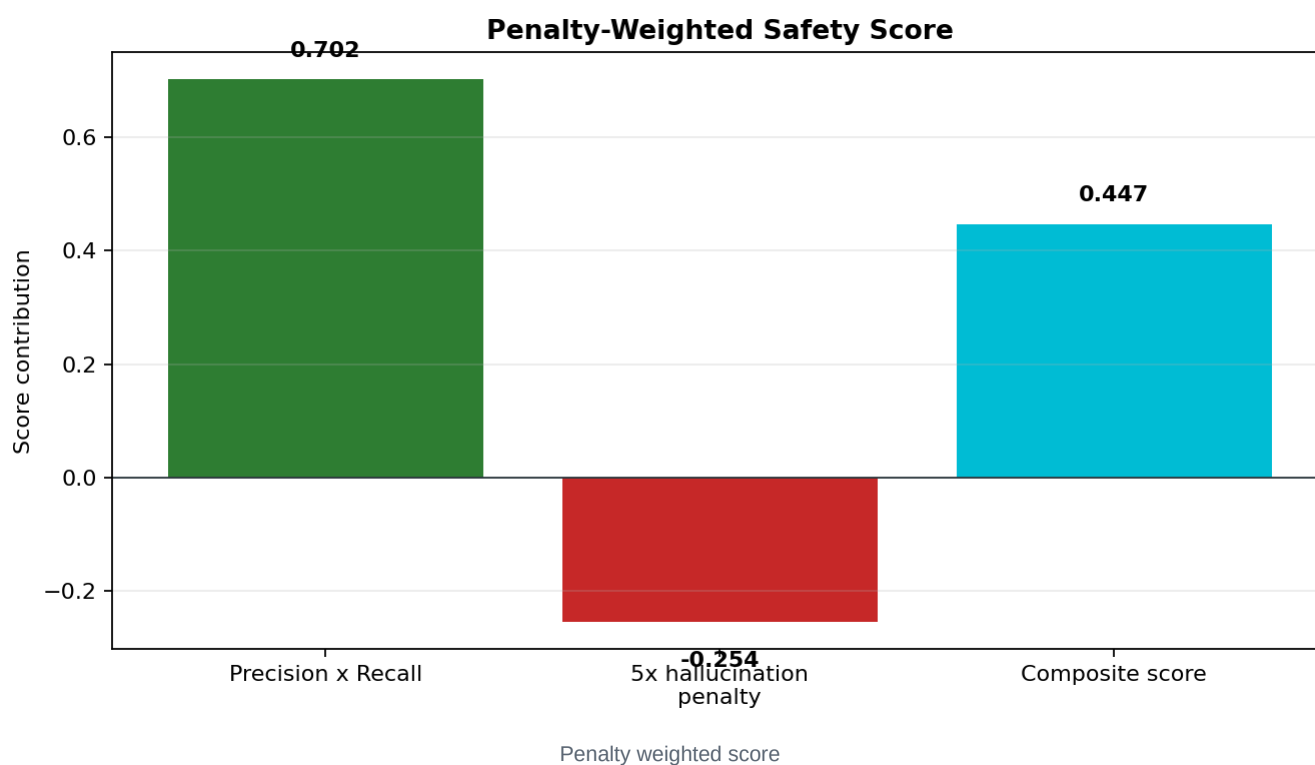
Penalty-Weighted Scoring

The sample report uses the implemented score form:

$$\text{Composite Score} = (\text{Precision} * \text{Recall}) - (5 * \text{Hallucination Rate})$$

The 5x penalty makes unsupported clinical assertions more costly than ordinary text overlap errors.

Component	Value
Precision	0.885
Recall	0.793
Precision x Recall	0.702
Hallucination rate	0.051
5x hallucination penalty	0.254
Composite score	0.447



Claim Verification Flow

Claim Verification Flow

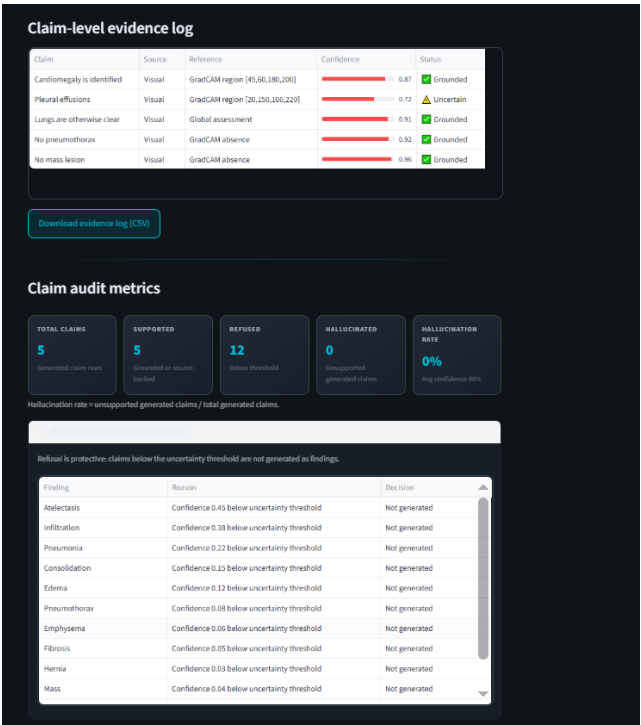


Audit principle: every generated claim should have evidence, a confidence score, and a reviewable decision.

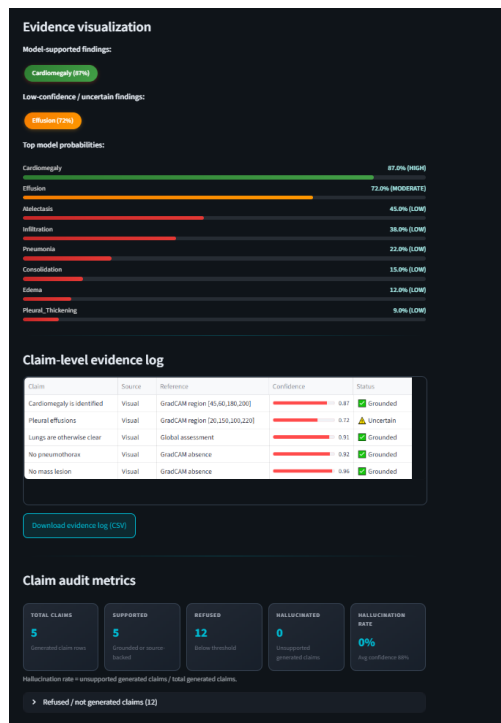
Claim audit flowchart

Dashboard Evidence

The dashboard exposes the same evidence trail interactively.



Evidence log table from README



Evidence visualization from README

Reproduce

Regenerate the evidence log, plots, Markdown reports, and PDFs:

```
python scripts/generate_report_artifacts.py
```

For a real benchmark, run `scripts/evaluate.py` after configuring approved datasets and model checkpoints. Replace the sample audit with the exported benchmark rows before making clinical performance claims.

Limitations

- This report is an offline sample audit and should not be described as a clinical benchmark.
- The bundled sample cases are intended to prove pipeline behavior and deliverable format.
- Protected medical datasets require approval and local credentials.
- Grad-CAM boxes explain model attention but do not equal ground-truth anatomic annotations.